

[Resources](#) / [Homework/Quiz questions \(/COMP3331/23T1/resources/83523\)](#)

/ [Students to Discuss Solution: Headers, Headers and Headers \(Week 2\)](#)

Students to Discuss Solution: Headers, Headers and Headers (Week 2)

Consider a 4-layer protocol stack with the following layers (numbered 1 to 4 from bottom to top), physical, network, transport and application. The physical layer uses one-byte flags to mark both the start and end of a packet (i.e. a one-byte header and a one-byte trailer). The other three layers all use four-byte headers. The transport layer imposes that the maximum packet payload (i.e. data portion) for this layer must not exceed 100 bytes. Other layers have no such restrictions. Calculate the percentage overhead assuming an application message of 450 bytes is to be transmitted.

HINT: The application layer message cannot fit inside a single transport layer segment (450 bytes > 100 bytes) and will have to be divided up into multiple segments.

Resource created 2 months ago (Tuesday 17 January 2023, 01:58:45 PM), last modified about a month ago (Wednesday 08 February 2023, 05:15:27 PM).